

Ahmed Ikram | AI Engineering

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PROFILE

4th-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Builds production AI systems across agentic LLM applications, applied NLP, model fine-tuning, execution-based evaluation, and MLOps infrastructure - including a SageMaker-deployed anomaly detection engine achieving 99.8% cross-validated accuracy and an LLMops platform that improved Qwen3-14B SWE-bench fail-to-pass performance 7.0× over the base model. Seeking a graduate AI Engineering role focused on reliable model development, evaluation, and deployment.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

PROJECTS

ATM Log Aggregation, Analysis & Diagnostics Platform | *FastAPI, XGBoost, Isolation Forest, MLflow, SageMaker,*

LangChain, ChromaDB, Kafka, Redis, Docker, AWS

- Backend and ML engineering lead on a 7-person Agile team for NCR Atleos; independently built the 3-layer detection engine, Agentic RAG assistant, full MLOps infrastructure, AWS SageMaker production inference endpoint, and a 1,402-test suite - all post-deadline on top of the team delivered foundation
- 3-layer anomaly detection engine runs every 30s ML_Ensemble (XGBoost + Isolation Forest) on 49 hand-engineered features across 7 groups achieving 99.8% ±0.1% StratifiedKFold CV accuracy across 8 classes and 97.3% IF precision via grid search and Youden's J threshold calibration (F1=0.7008 at -0.51009); ZScore detects distribution shift via rolling 20-window Z-scores independently cross-checked against a deployed SageMaker endpoint (laad-xgb-champion, ml.t2.medium) for cloud-side validation without making inference dependent on cloud availability; all runs and 7 artifacts versioned in MLflow (RDS backend + S3 artifact store with champion aliases)
- Agentic RAG diagnostic assistant built over ChromaDB + Ollama with cross-encoder reranking (+5–15% retrieval relevance over bi-encoder cosine), self-consistency scoring (3 parallel LLM samples via ThreadPoolExecutor, 3-gram Jaccard similarity), reflexion self-critique (generate → critique → regenerate), and 4-signal confidence fusion (retrieval 30% + consistency 25% + verbalized 25% + grounding 20%) with Platt calibration on ECE trigger; parallelising self-consistency cut end-to-end latency from 26–53s to 11–23s; responses served at <100ms from Redis cache
- 1,402 tests (923 pytest backend at 82% coverage across 10 tiers, 394 vitest frontend, 10 Playwright E2E, 75 Terraform assertions) gating every PR - catching 7 critical defects including a JWT privilege escalation vulnerability and silent data loss under 50 concurrent writes; full AWS estate provisioned via Terraform (10 modules, 118 resources) with 3 GitHub Actions pipelines (OIDC auth, zero static credentials)

StockLens FinTech Spending & Investment App | *Python, FastAPI, Airflow, PostgreSQL, Redis, MLflow, LangGraph, Optuna,*

PyTorch, Rust, Terraform, AWS

- LangGraph ReAct financial assistant agent exposing 16 tools across 7 categories to AWS Bedrock Converse with SSE streaming responses, and Elasticache/RDS conversation persistence that survives server restarts with full history reconstruction, and LLM-as-judge evaluation with LangSmith tracing
- Confidence-gated 4-stage OCR and NLP extraction pipeline for receipt understanding, combining Tesseract Regex, Bedrock Vision, text-only LLM, and merchant/category resolution with RapidFuzz matching and a 24h Redis cache keyed on receipt hash avoids repeat inference on duplicate scans
- Weekly automated retraining and production ML operations via Airflow on ECS Fargate: 6-year OHLCV ingestion, feature generation, MLflow-tracked training, champion/challenger, promotion gated at >2pp directional accuracy improvement with delivery to EFS, S3, and a PostgreSQL model registry: Evidently AI drift monitoring via PSI/KS/JSD thresholds with alerting to CloudWatch and reports archived to S3
- Trained a global LSTM for 3-class price-direction forecasting across the S&P 500 tickers, achieving 51.6% directional accuracy against a 33% random baseline and a 0.97 simulated Sharpe ratio; accelerated training and live inference with a custom Rust/PyO3 feature engine replacing pandas-based computation

- Terraform IaC (14 modules, 193 resources) on ECS Fargate ARM64/Graviton behind a WAF-protected ALB, with Multi-AZ RDS, ElastiCache Redis, and CPU/RPS-based autoscaling; validated via 1,415 backend tests and 822 frontend assertions at 90% coverage gates, enforced by a 9-parallel-job CI pipeline and a 7-stage OIDC-authenticated CD pipeline

SWE-Owen Automated LLM Fine Tuning, Evaluation & Serving Platform | *Python, PyTorch, QLoRA, Unsloth, Hugging Face, Modal, vLLM, FastAPI, Terraform, Google Cloud, Weights & Biases, GitHub Actions*

- A model-agnostic LLMOps platform spanning schema-validated data preparation, QLoRA fine-tuning, execution-based evaluation, statistical champion-challenger promotion, and OpenAI-compatible inference serving, with configuration-driven model selection and versioned GCS/W&B artifacts shared across training, evaluation and serving
Built a Six-stage data pipeline processing 20,477 SWE-bench issues into a 17,456-example high-signal training corpus: Pydantic schema validation, 6 automated quality gates, exact/semantic deduplication, and per-repo stratified 80/10/10 splitting across 46 repos to eliminate cross-repo leakage - yielding 14,833 tokenized training examples and a 2,313-instance golden evaluation set before tokenization
- Fine-tuned 3 QLoRA variants of Qwen3-14B using 4-bit NF4 quantization via Unsloth + FlashAttention on Modal A100-80GB GPUs, 1.76% of parameters trained and built an execution-based evaluation harness that materializes each repo inside its official SWE-bench Docker image and runs real FAIL_TO_PASS/PASS_TO_PASS test suites; the promoted champion scored 17.20% F2P (95% Wilson CI 11.1–25.8%, 7.0× improvement, McNemar p<1e-6) at 90.10% P2P, with promotion gated behind a 4-condition statistical test (absolute floors, CI lower bound, and a P2P-drop ceiling)
- Deployed an OpenAI-compatible inference API (FastAPI, SSE streaming) on Modal scale-to-zero infrastructure with per-request LoRA adapter swapping via vLLM + AWQ 4-bit quantization (zero engine restarts, \$0 idle cost) provisioned entirely via Terraform with GCP Workload Identity Federation (OIDC, zero static credentials); validated via 4 GitHub Actions workflows, including a per-PR 20-instance execution-based smoke gate where only main-branch runs can write the regression baseline, making self-certified regressions impossible, and 1,456 tests with strict mypy typing

TECHNICAL SKILLS

Languages	Python · SQL · TypeScript · JavaScript · Bash · Rust
LLM & Generative AI	PyTorch · Hugging Face Transformers · QLoRA · LoRA · PEFT · TRL · Unsloth · AWS Bedrock
AI Systems & Evaluation	LangGraph · LangChain · RAG · ChromaDB · Ollama · LLM evaluation · SWE-bench · Tesseract OCR · RapidFuzz
MLOps & Model Serving	MLflow · Airflow · vLLM · FastAPI · Evidently AI · Weights & Biases · LangSmith · Langfuse
Data & Backend	Kafka · PostgreSQL · Redis · MySQL · MongoDB · Pandas · NumPy · Pydantic
Cloud, Infrastructure & Testing	AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, IAM, SageMaker, Bedrock, CloudFront, CloudWatch, Secrets Manager) · Google Cloud · Modal · Terraform · Docker · GitHub Actions · OIDC · PyO3 · Maturin · Pytest · mypy · Ruff

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>

EXPERIENCE

Beaully Mini Market , Dundee, Scotland	
<i>Retail Assistant</i>	<i>September 2023 - Present</i>
• Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption	
Zulus Peri Peri Grill , Dundee, Scotland	
<i>Front of House Worker</i>	<i>June 2021 - September 2023</i>
• Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure	